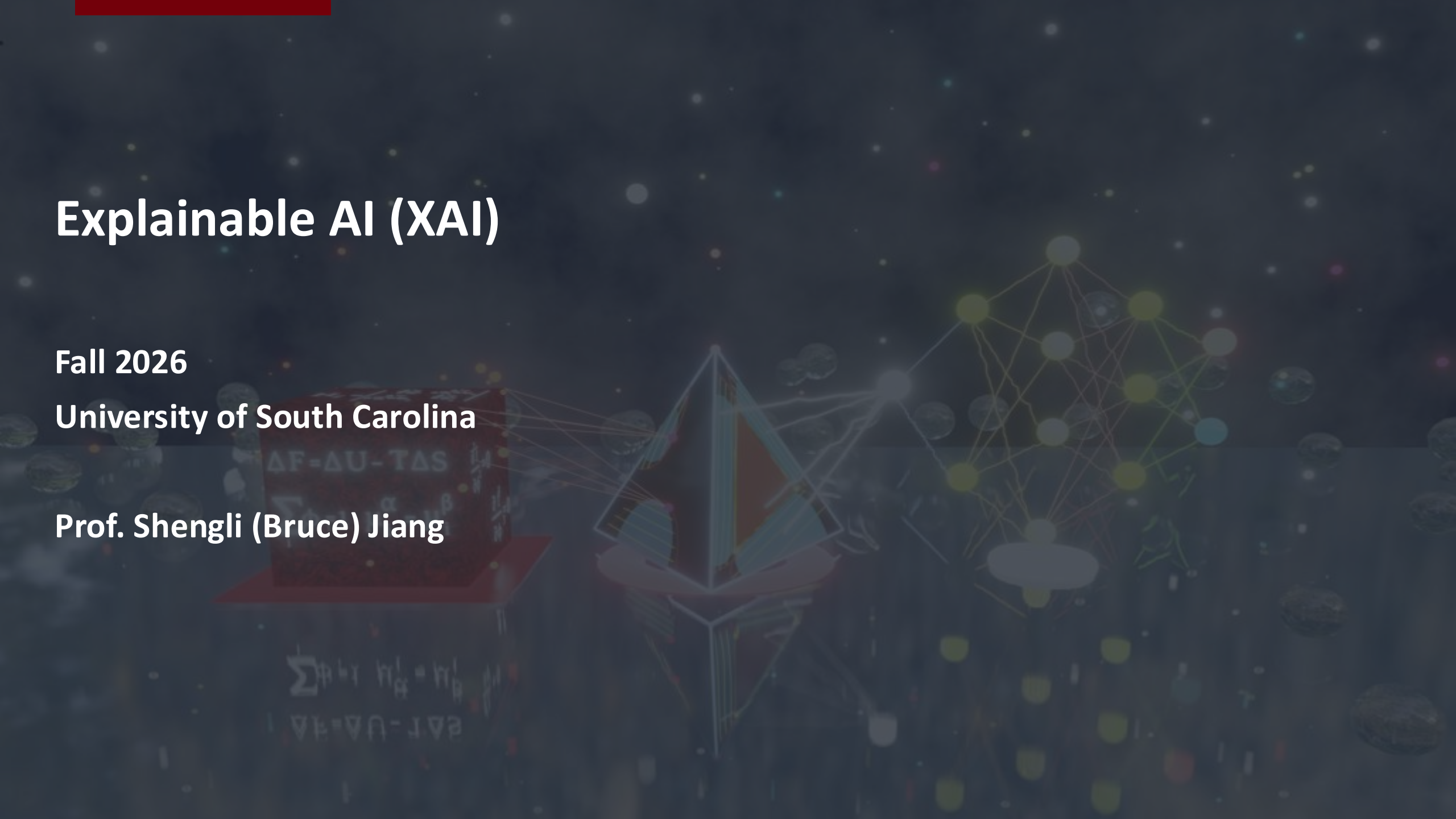
Explainable AI (XAI)

Fall 2026

University of South Carolina

Prof. Shengli (Bruce) Jiang



From Lecture 22

Unsupervised Learning III: Clustering

- *Partition-based clustering with K-means and K-medoids*
- *Choosing the number of clusters using elbow and silhouette methods*
- *Density-based clustering with DBSCAN*

In this lecture, you will learn

- *Identify influential features using permutation importance*
- *Interpret feature effects using partial dependence plots (PDP)*
- *Explain individual predictions using SHAP analysis*

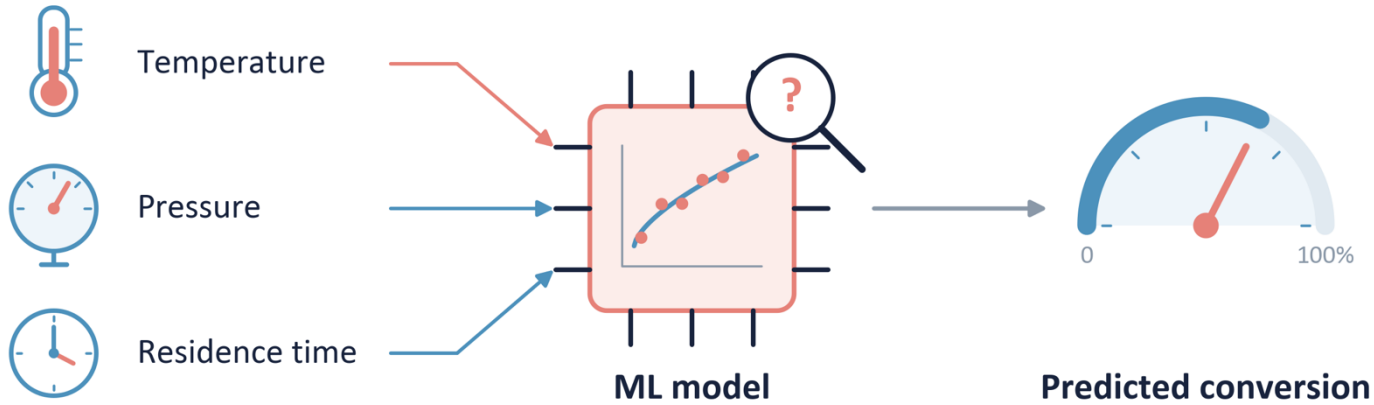
Motivation for XAI

Explainable AI helps us understand how a model uses inputs to make predictions.

Understand reliance
- Which inputs matter most?

Check reliability
- Is the model using reasonable information?

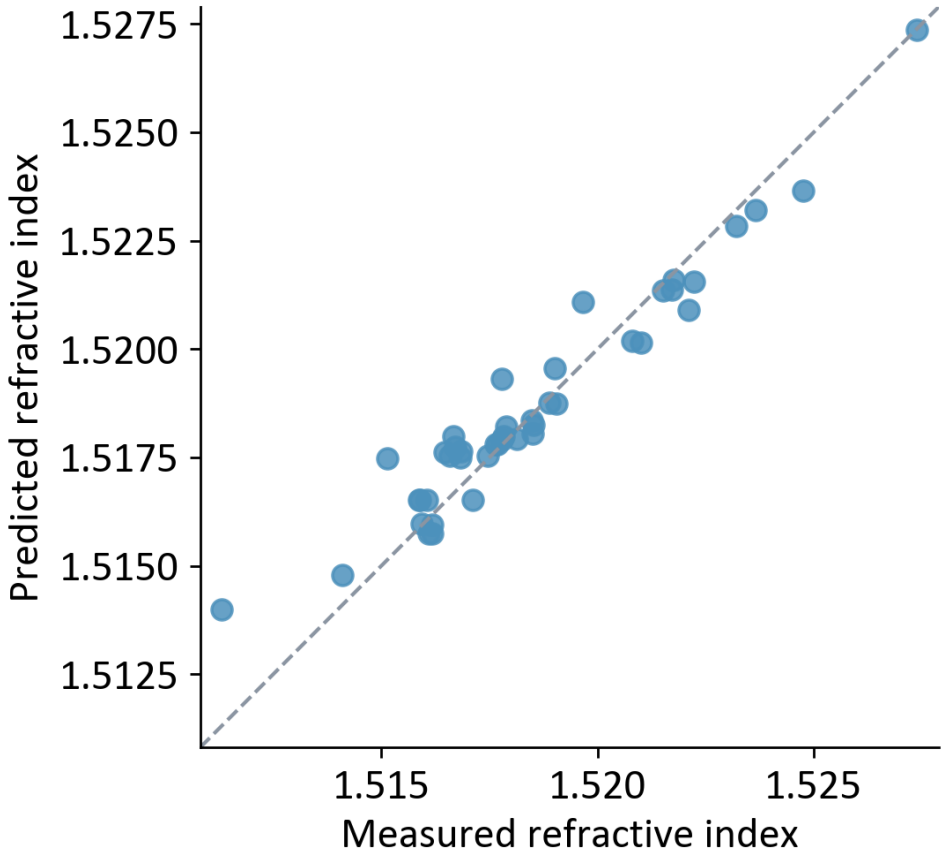
Guide investigation
- Which inputs deserve further study?



Which input would most affect the model if its information were disrupted?

Glass Regression

The regression predicts refractive index from eight measured composition variables: Na, Mg, Al, Si, K, Ca, Ba, and Fe.



- 214 samples, eight composition features.
- Ridge regression, evaluated on 43 unseen samples.
- Test RMSE = 0.00085.

We have a model that predicts well. Now we want to understand what it relies on.

Which composition variable matters most to this prediction model?

Feature Shuffling

Original test data			Shuffled test data		
Sample	Na	**Ca**	Sample	Na	**Ca**
205	14.95	**8.71**	205	14.95	**9.85**
51	13.72	**10.06**	51	13.72	**8.71**
128	13.78	**9.85**	128	13.78	**10.06**

Only Ca values are rearranged.
Other features, measured refractive indices, and the fitted model remain unchanged.

If the model relies strongly on Ca, what happens to prediction error after shuffling it?

Permutation Importance

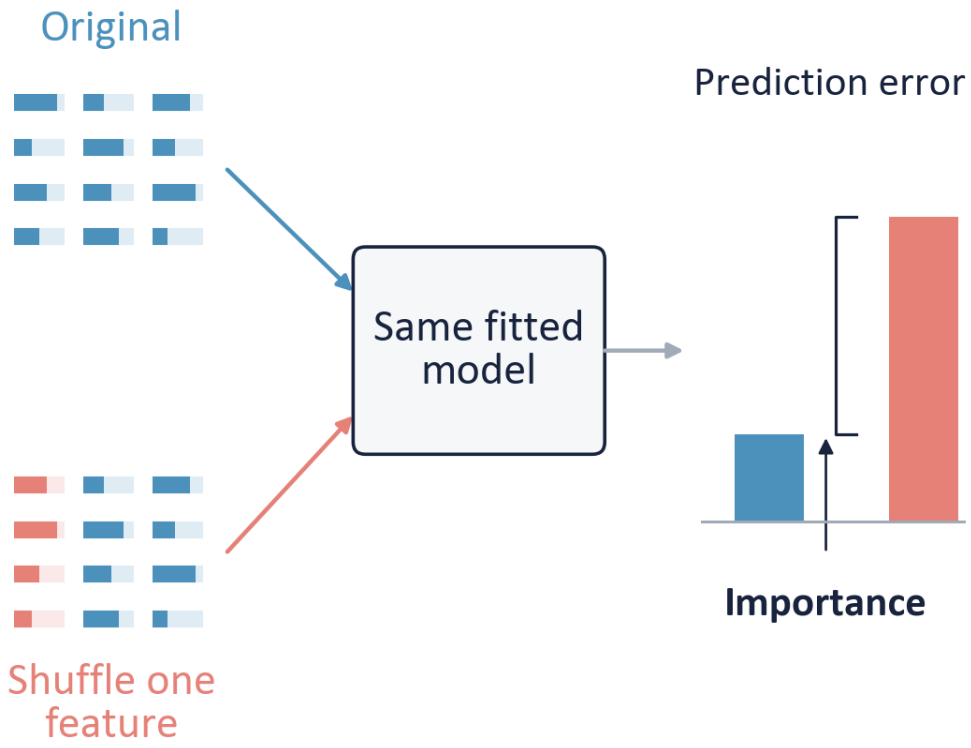
Permutation importance measures how much a fitted model's error increases when one feature is randomly shuffled.

Algorithm

1. Measure the original test RMSE.
2. Shuffle one feature and recompute RMSE.
3. Repeat and average the increase in RMSE.

$$I_j = \frac{1}{K}\sum_{k=1}^{K}(E_{k,j} - E_0)$$

- I_j = importance of feature j
- E_0 = original test RMSE
- $E_{k,j}$ = RMSE after the k-th shuffle of feature j
- K = number of repeated shuffles

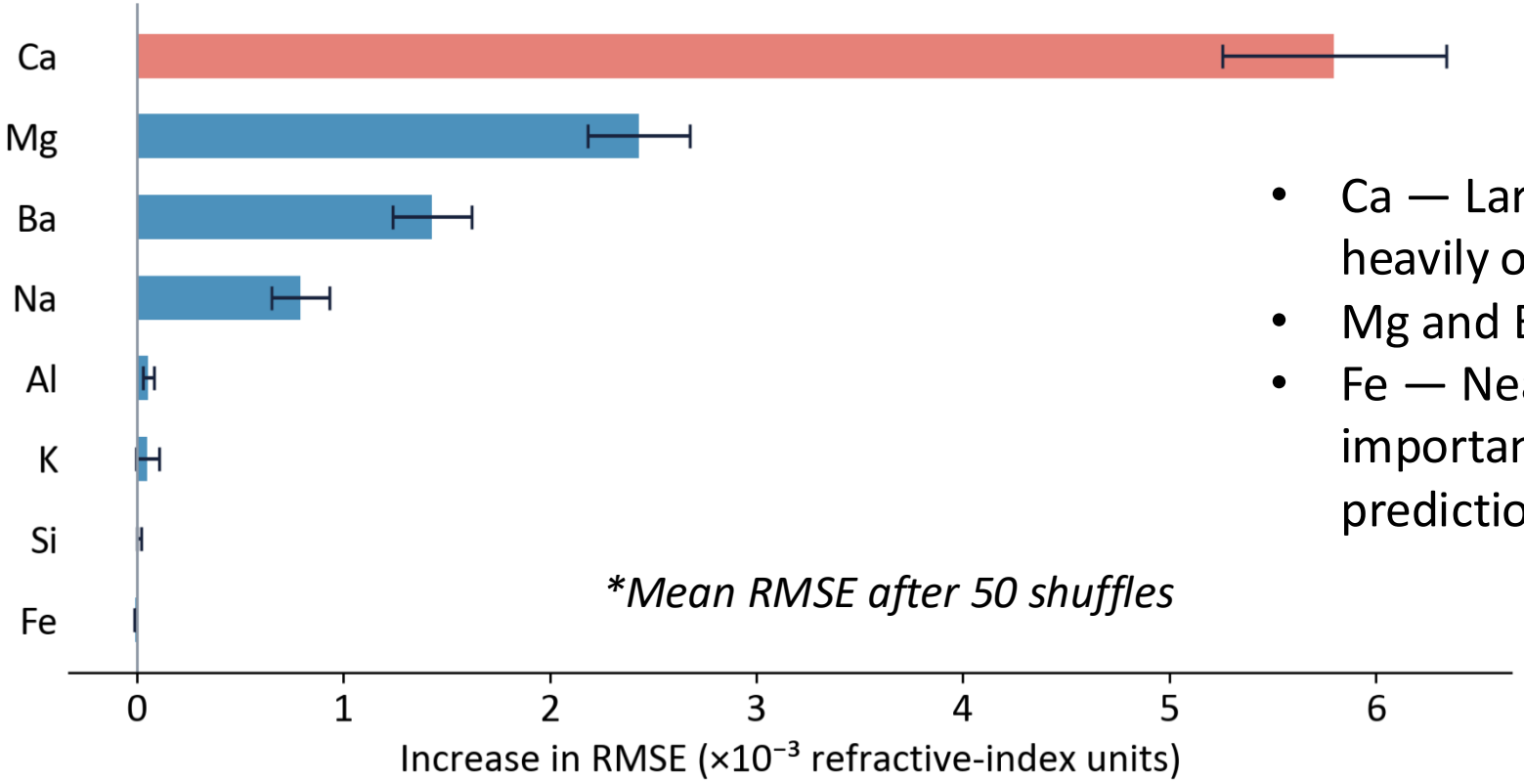


Shuffle the test data, not the trained model. Do not retrain.

Permutation Results

What happens when we shuffle Ca?

Original RMSE	After shuffling	Importance
0.00085	0.00665	+0.00580



- Ca — Large increase in error. The model relies heavily on Ca.
- Mg and Ba — Also important.
- Fe — Near-zero or slightly negative importance. Shuffling Fe does not hurt predictions.

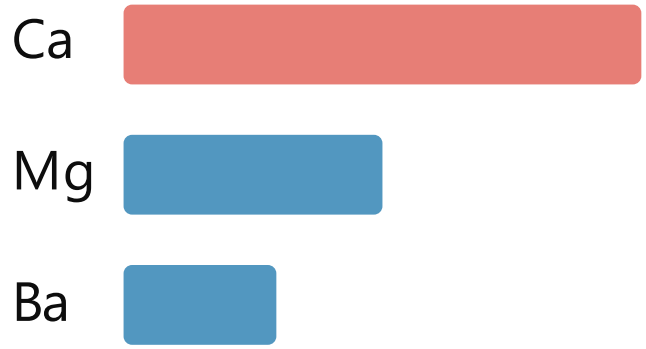
Limitations of Permutation Importance

Permutation importance tells us	It does not tell us
Which features the model relies on	Whether a feature causes the outcome
How strongly shuffling hurts accuracy	Whether increasing a feature increases the prediction
Importance for a particular model and dataset	Whether the learned relationship is physically correct

Feature Effects

Permutation importance tells us which features the model relies on. But it does not tell us how predictions change as a feature varies.

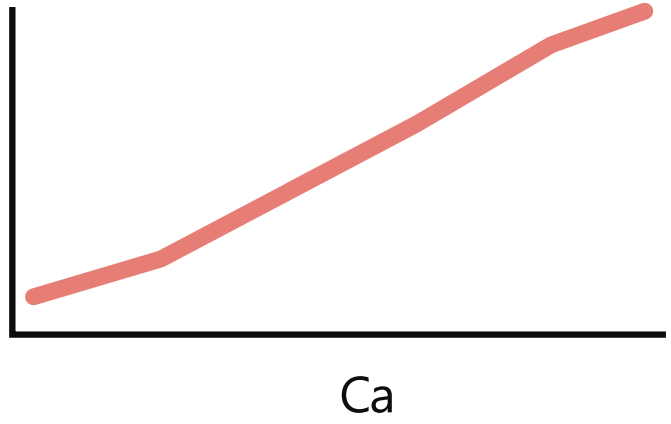
Permutation importance



Which feature matters?

Ca is important.

Partial dependence



How does prediction change?

Higher Ca gives higher predictions in this model.

PDP Calculation

Set Ca = 8 for every test sample

Sample ID	Original Ca	Ca set to	Na	Mg	New Predictions
205	8.71	**8.00**	14.95	0.00	**1.51376**
51	10.06	**8.00**	13.72	3.72	**1.51708**
128	9.85	**8.00**	13.78	2.28	**1.51502**

If all three glasses now have Ca = 8, why are their predictions different?

Their other seven composition features remain different.

Average predictions over all 43 samples: 1.51600

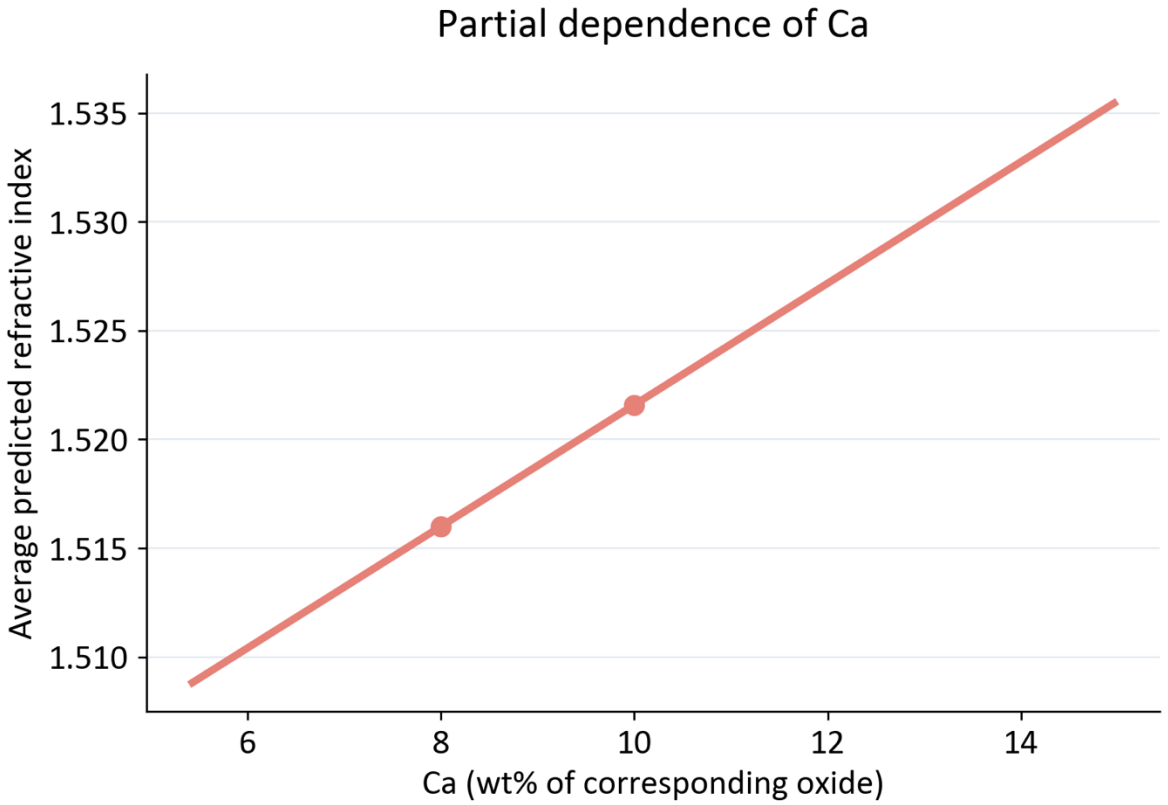
PDP Construction

Set Ca = 8 for every test sample → average prediction 1.51600

Set Ca = 10 for every test sample → average prediction 1.52159

Difference in average prediction: 1.52159-1.51600 = 0.00559

Repeat for many Ca values, then connect the resulting points to form the PDP.



Conclusion: The fitted Ridge model predicts a higher average refractive index as Ca increases.

Since Ridge regression is linear in the standardized inputs, the resulting PDP is straight.

Partial Dependence

A partial dependence plot shows the average model prediction as one feature is set to different values, averaging over the observed values of the other features.

Algorithm

1. Choose a feature and a set of values.
2. Assign each value to all samples, keeping other features unchanged.
3. Predict, average, and plot.

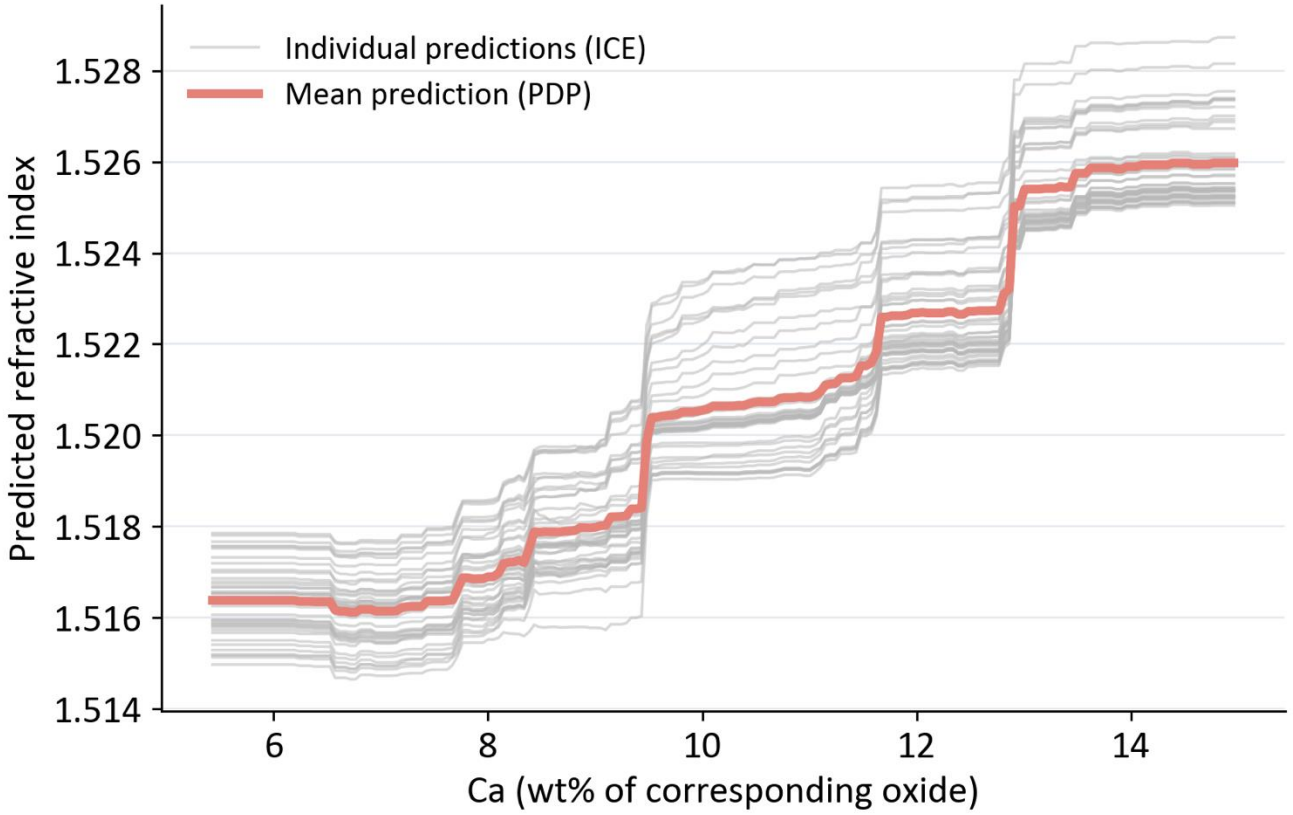
$$\widehat{\mathrm{PD}}_j(z) = \frac{1}{n}\sum_{i=1}^{n} \hat{f}(z, \mathbf{x}_{i,-j})$$

- z: chosen value of feature j
- $\boldsymbol{x}_{i,-j}$: sample i's other inputs
- $\hat{f}$: fitted prediction model
- n: number of samples used for averaging

- *A useful contrast with permutation importance is that PDP **does not** require measured target values.*
- *It averages predictions, whereas permutation importance calculates changes in a performance metric using measured targets.*

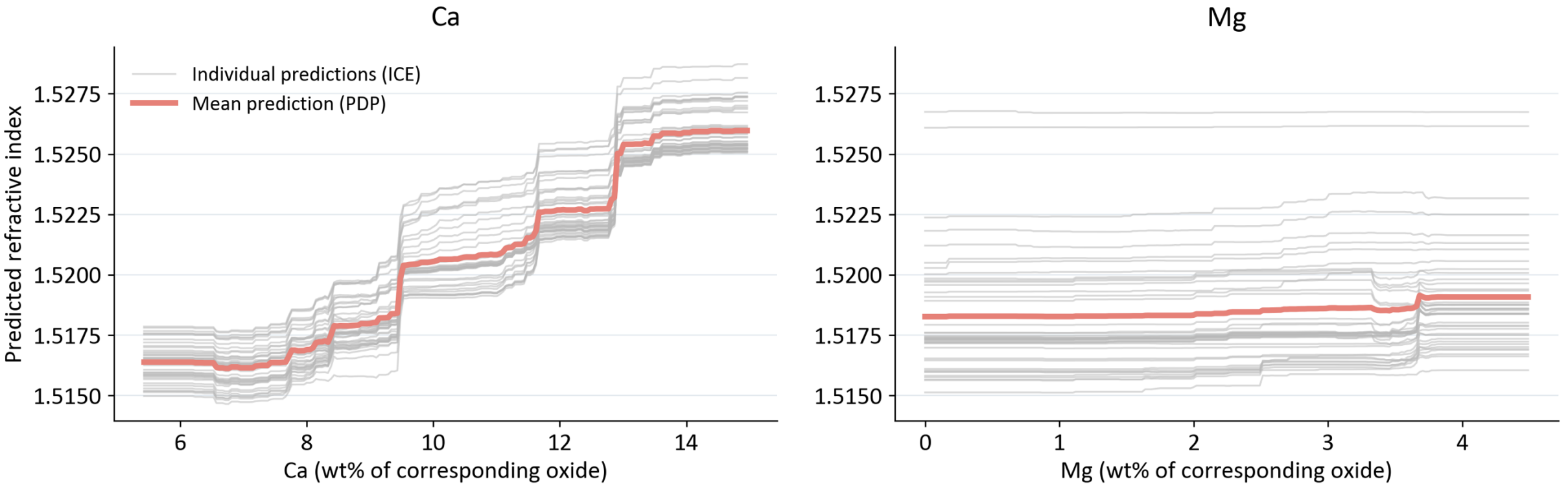
Nonlinear PDP

Example: Random forest regression predicts refractive index from the same eight glass composition variables.



- The random forest predicts a ***nonlinear response***, with *plateaus* and *jumps* caused by tree splits.
- Each gray curve follows one sample as Ca varies. The red PDP is their average at each Ca value.

Comparing Ca and Mg



Ca: Large, nonlinear increase in average predicted refractive index.

Mg: Much smaller change over the displayed range.

A PDP describes the model's response to hypothetical inputs, not necessarily a physically realizable or causal effect.

Introduction to SHAP

Permutation importance	Which features matter for predictive performance?
PDP	How does the average prediction change with Ca?
SHAP	Why is this particular glass prediction relatively high?

- Reference model prediction: **1.51833**
- Prediction for sample 51: **1.52341**

Which features explain the difference of +0.00508?

SHAP Decomposition

Prediction decomposition

$$\widehat{f}(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j(\mathbf{x})$$

- ϕ_0: reference prediction
- $\phi_j(\boldsymbol{x})$: contribution assigned to feature j for this sample
- p: number of input features

For sample 51: $1.52341 = 1.51833 + 0.00508$

- The reference is the average model prediction over the chosen *background dataset*.
- Each feature receives a contribution relative to that reference.
- The contributions sum to the difference between prediction and reference.

$$\phi_0 = \frac{1}{171} \sum_{b=1}^{171} \widehat{f}(\mathbf{b}_b) \approx 1.51833$$

Here, our background contains 171 training samples. We use their prediction average as its baseline.

Feature Coalitions

Suppose the model gives these averaged predictions when features are *included*

Features fixed to target sample	Average prediction
Neither Ca nor Si	1.500
Ca only	1.510
Si only	1.504
Both Ca and Si	1.520

- Here, *included* means fixing that feature to the sample's observed value.
- *Missing features* are filled using background data, and predictions are averaged.

How much did Ca contribute?

1.510-1.500=0.010

Feature Ordering

Order A — Include Ca first

Neither	+ Ca	+ Si
1.500 reference	1.510 +0.010 for Ca	1.520 +0.010 for Si

Order B — Include Si first

Neither	+ Si	+ Ca
1.500 reference	1.504 +0.004 for Si	1.520 +0.016 for Ca

Ca contributes +0.010 when added first, but +0.016 when added after Si.
Why? The model's response to Ca depends on which other information is already included.

Which contribution should we assign to Ca?

Shapley Values

$$\phi_{\mathrm{Ca}} = \frac{0.010 + 0.016}{2} = 0.013$$

$$\phi_{\mathrm{Si}} = \frac{0.004 + 0.010}{2} = 0.007$$

$$\underbrace{1.500}_{\text{reference}} + \underbrace{0.013}_{\text{Ca}} + \underbrace{0.007}_{\text{Si}} = \underbrace{1.520}_{\text{prediction}}$$

SHAP assigns each feature its average contribution across possible feature orders.

SHAP Calculation

SHAP explains an individual model prediction by assigning each feature a contribution relative to a reference prediction.

Algorithm

- Choose a sample and reference dataset.
- Compare predictions with and without one feature.
- Average the changes across feature orders.
- Repeat for every feature.

Marginal contribution

$$\Delta_j(S) = v(S \cup \{j\}) - v(S)$$

- S: a subset of features already included
- j: the feature being evaluated
- $v(S)$ is the average model prediction when the features in S are fixed to the target sample and the remaining inputs are taken from background data.

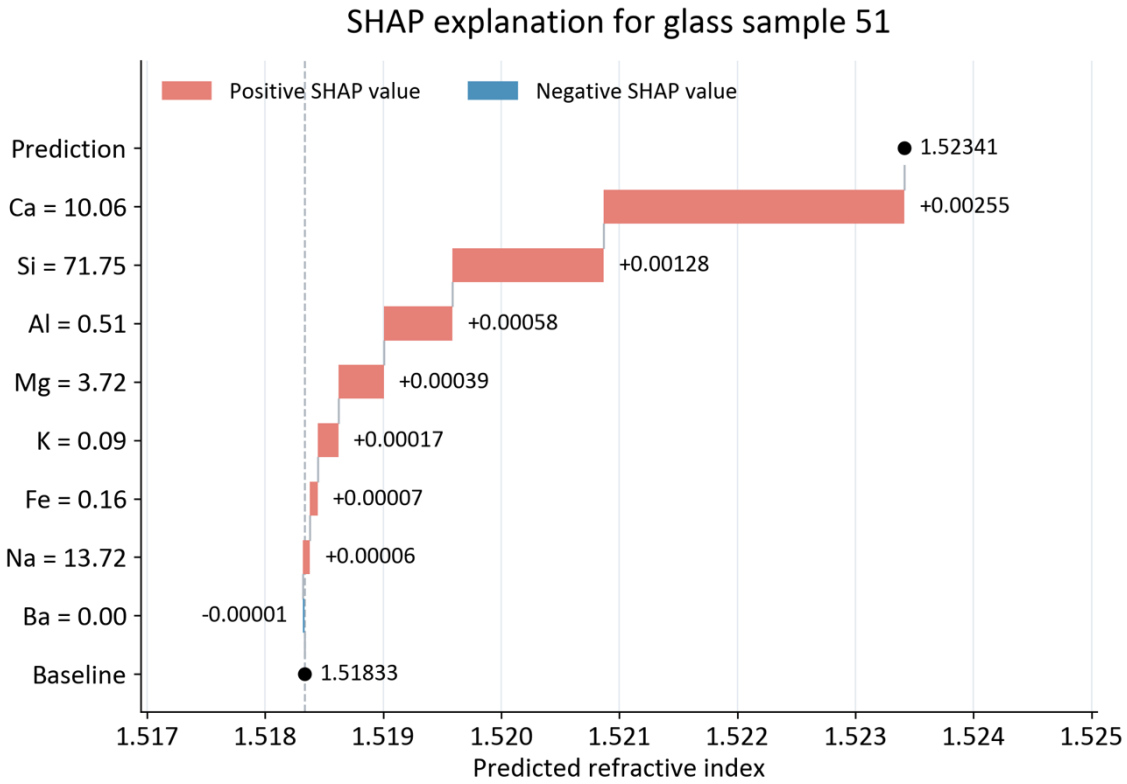
The SHAP value averages $\Delta_j(S)$ across all feature orders.

Local SHAP

Example: glass sample 51, explained using the fitted random forest.

$$\phi_0 = \frac{1}{171}\sum_{b=1}^{171}\widehat{f}(\mathbf{b}_b) \approx 1.51833$$

$$\underbrace{1.52341}_{\text{prediction}} \approx \underbrace{1.51833}_{\text{reference}} + \underbrace{0.00508}_{\text{total SHAP contribution}}$$



Reading the waterfall plot

- Begin at the reference prediction.
- Red bars show positive contributions.
- Blue bars show negative contributions.
- Adding all eight contributions reaches the model prediction.

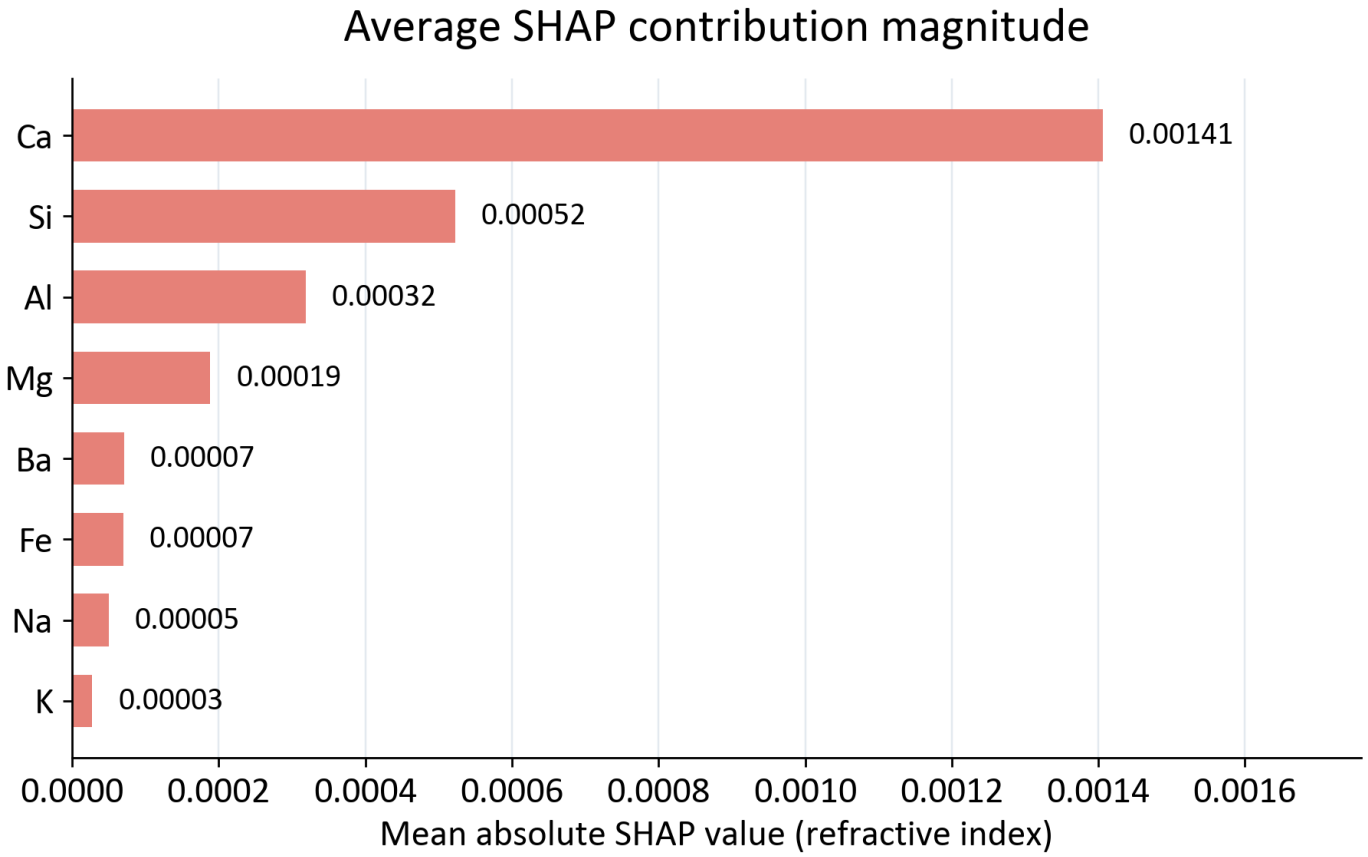
Does this mean Ca is always the most important feature for every glass sample?

No. SHAP values are sample-specific.

For this sample

- Ca contributes approximately (+0.00255).
- Si contributes approximately (+0.00128).
- Ca and Si provide the largest positive contributions.

Global SHAP



$$I_j^{\mathrm{SHAP}} = \frac{1}{n}\sum_{i=1}^{n} |\phi_j(\mathbf{x}_i)|$$

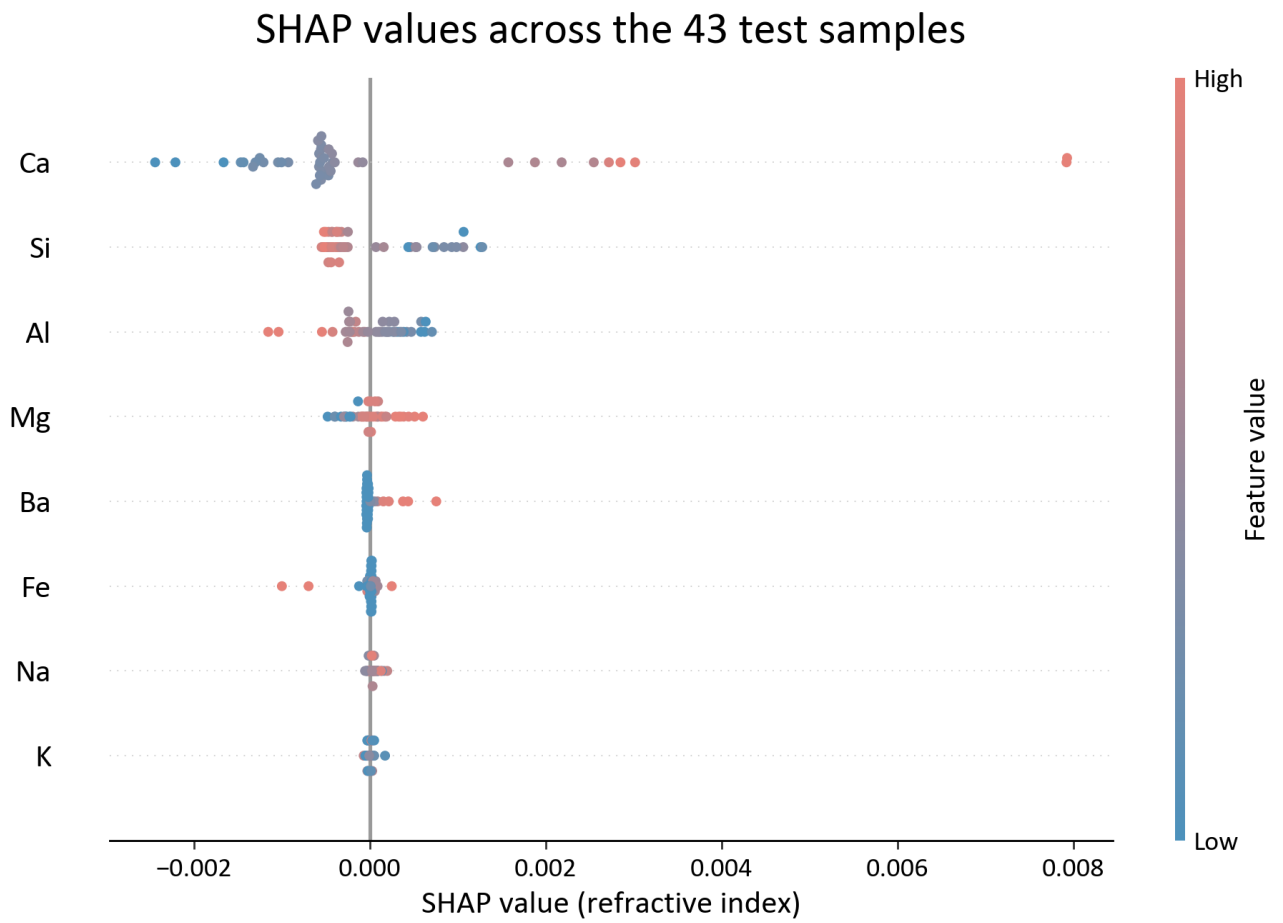
Here, n is the number of samples being summarized and $\boldsymbol{x}_i$ is sample i.

Conclusion: Ca has the largest mean absolute SHAP value, approximately 0.00141.

SHAP Beeswarm

How do contributions vary across individual glass samples?

Visual element	Meaning
Each dot	One sample's SHAP value
Horizontal position	Signed contribution
Red dot	High feature value
Blue dot	Low feature value
Row	One feature



Example: Ca shows the largest overall SHAP contributions, indicating that the random forest relies strongly on this feature. High Ca values (red dots) generally have positive SHAP values, pushing predicted refractive index above the reference prediction, while low Ca values (blue dots) tend to push predictions downward. The wide horizontal spread indicates that Ca's contribution varies substantially across glass samples, with some positive contributions reaching approximately +0.008.

Comparing XAI Methods

Method	What we learn
Permutation importance	How much shuffling a feature hurts predictive performance
PDP	How the average model prediction varies with a feature
SHAP	How features contribute to predictions relative to a reference
Random forest impurity importance	How features contributed to reducing split impurity during training

Model explanations are not necessarily physical mechanisms. SHAP describes how the fitted model assigns contributions, not whether Ca causes refractive index to increase.

Interactive Python Time

23_XAI.ipynb

In this lecture, we learned

- *Identify influential features using permutation importance*
- *Interpret feature effects using partial dependence plots (PDP)*
- *Explain individual predictions using SHAP analysis*

Next lecture:

- *Molecular and Materials Featurization*
  - *Tokenization and One-Hot Encoding*
  - *Fingerprints*
  - *Molecular Descriptors*